

Inverting Buck-Boost

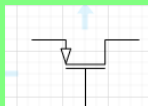


J1
108-0906-001
+24V



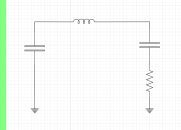
J4
108-0903-001
GND

Input switch

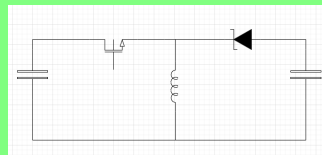


UVLO/OVLO

Input filter



Power stage



PWM

CSP

CSN

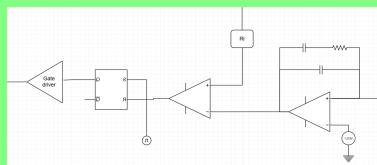
CSP

CSN

Current sense

CS

Control circuit



PWM

OCF

RAMP

FB

-12V



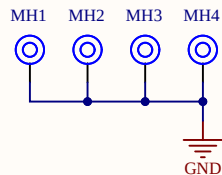
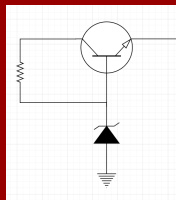
J2
108-0906-001



J3
108-0903-001

GND

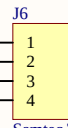
Linear regulators



J5
Signal
GND
SHIELD
SHIELD
031-5431-10RFX

1
2
P1
P2
GND

-12V



Samtec TS-104-T-AA

J6 is a 4-pin header that selects the op-amp negative input source: either the injected sawtooth signal (to subtract it from the compensator output) or -12 V to disable this feature.

J5 is a female BNC connector used to inject an external compensating ramp (sawtooth) if needed

-12V

R1
1.13k
0.1%

R2
10k
0.1%

GND

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Power stage

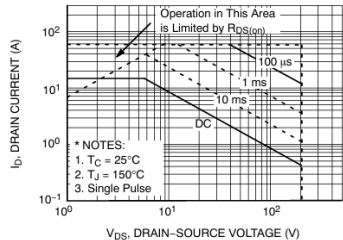
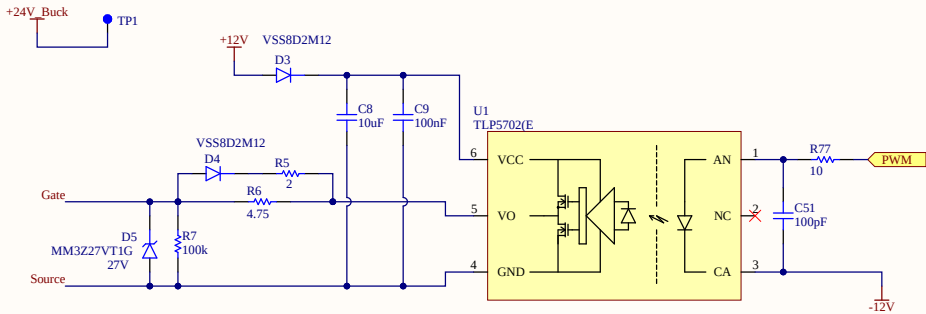
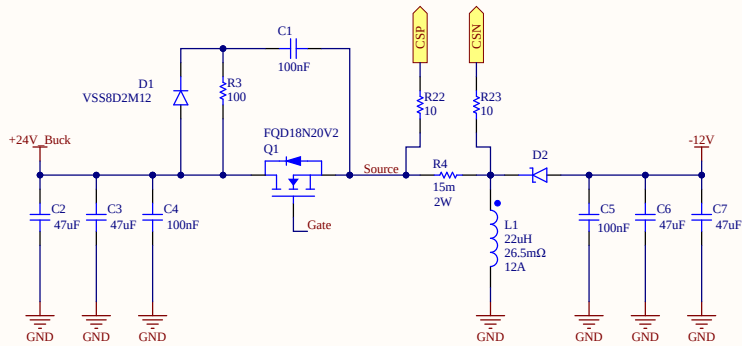


Figure 9. Maximum Safe Operating Area

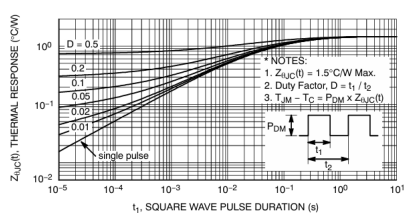


Figure 11. Transient Thermal Response Curve

Latch-off / hiccup OCP: Q1 hits 11A once, OCP trips, gate goes off, converter stops (or waits to retry). The 396W linear event happens once → single pulse is correct.
Cycle-by-cycle OCP: Q1 hits 11A, gets turned off for that cycle, then turns back on next cycle and hits 11A again... repeating every switching period for several ms during a sustained overload. Now the high-power linear crossing repeats every cycle → this is a pulse train, and you should use the duty-cycle curve, NOT single pulse.

FQD18N20V2 [Q1] — SOA CALCULATION

Package: DPAK | 200V | 15A | $R_{th(j-c)} = 1.5^{\circ}\text{C/W}$ | $T_{jmax} = 150^{\circ}\text{C}$
 $T_A = 40^{\circ}\text{C}$ (max ambient)
SOA points (Fig.9 @ $V_{ds} = 36\text{V}$, 25°C):
 $t_1 = 1\text{ms} \rightarrow \text{ISOA} = 20\text{A}$
 $t_2 = 10\text{ms} \rightarrow \text{ISOA} = 7\text{A}$
POWER-LAW FIT:
 $\text{SOA}(t) = a \times t^m$ (t in seconds)
 $m = \ln(20/7) / \ln(1\text{ms}/10\text{ms}) = 1.050 / -2.303 = -0.456$
 $a = 20(1\text{ms})^{(-0.456)} = 0.8575$
 $\text{SOA}(t) = 0.8575 \times t^{(-0.456)}$

CASE 1 — NORMAL SWITCHING

- ON-state V_{ds} in $R_{DS(on)}$ region, SOA-safe (low V_{ds})
- Switching edges: cross active region but last ~20ms < 10μs → far inside SOA pulse limit
- $\text{ID}_{peak} = 5.31\text{A} < \text{SOA}(10\mu\text{s}) \rightarrow$ far inside SOA ✓

CASE 2 — STARTUP (OCP clamp = 11A)

$V_{ds} = \text{Vin} + |\text{Vout}| = 36\text{V}$
 $\text{P}_{pulse} = V_{ds} \times \text{I}_{OCP} = 36 \times 11 = 396\text{W}$
 $Z_{th(j-c)}$ read from Fig.11 SINGLE-PULSE curve:
 $t = 1\text{ms} \rightarrow Z_{th} = 0.17$ $t = 2\text{ms} \rightarrow Z_{th} = 0.25$ $t = 3\text{ms} \rightarrow Z_{th} = 0.30$ (arbitrary sweep points to locate the current/pulse limit)
 $T_j = T_A + Z_{th}(t) \times \text{P}_{pulse}$
 $\text{ISOA}_{derated} = \text{SOA}(t) \times (T_j - T_A) / (T_{jmax} - 25)$
 $t = 0.64\text{ms}$: $T_j = 40 + 0.136 \times 396 = 94^{\circ}\text{C}$
 $\text{SOA}_{25} = 0.8575 \times (0.64\text{ms})^{-0.456} = 24.5\text{A}$
 $\text{der} = 24.5 \times (150 - 94) / 125 = 11.0\text{A} < 11\text{A} \blacktriangleleft \text{LIMIT}$
 $t = 1\text{ms}$: $T_j = 40 + 0.17 \times 396 = 108^{\circ}\text{C}$
 $\text{SOA}_{25} = 20.0\text{A}$, $\text{der} = 20.0 \times (150 - 108) / 125 = 6.8\text{A} < 11\text{A} \times$

CHECK: $\text{I}_{OCP} = 11\text{A}$ vs $\text{ISOA}_{derated}$
 $t \leq 0.64\text{ms}$: $11\text{A} \leq \text{ISOA}_{derated}$ ✓ PASS
 $t > 0.64\text{ms}$: $11\text{A} > \text{ISOA}_{derated}$ ✗ FAIL

RESULT

Max safe linear pulse at 11A/36V ($T_A = 40^{\circ}\text{C}$) = 0.64ms
→ OCP must turn Q1 OFF within 0.64ms of reaching 11A.
Cycle-by-cycle: one period = 3.3μs @ 300kHz, far inside the 0.64ms safe window ✓

ANALYSIS

$\text{Vin} = 24\text{V}$ | $\text{Vout} = -12\text{V}$ | $\text{Iout} = 3\text{A}$ | $f_s = 300\text{kHz}$ | $D = 0.333$ | $\text{eff} = 88\%$

INDUCTOR — SRP1245A-220M (22uH)

$\text{IL}_{avg} = \text{Iin} + \text{Iout} = 1.705 + 3 = 4.705\text{A}$
 $\Delta\text{IL} = \text{Vin} \times D / (f_s \times L) = 1.21\text{A}$
 $\text{IL}_{peak} = \text{IL}_{avg} + \Delta\text{IL}/2 = 5.31\text{A}$
 $\text{IL}_{ripple_ratio} = 1.21 / 4.705 = 25.7\%$
 $L_{crit} = (D)^2 \times R / 2f_s = 2.97\mu\text{H}$
CCM confirmed: $L = 22\mu\text{H} \gg L_{crit} = 2.97\mu\text{H}$ ✓

MOSFET — FQD18N20V2 (200 V, 15 A, 140 mΩ)

$V_{ds_stress} = \text{Vin} + \text{Vout} = 24 + 12 = 36\text{V}$
 $V_{ds_rated} = 36 \times 1.2 = 43.2\text{V} \rightarrow 600\text{V}$ ✓ (20% derating)
 $\text{Id}_{rated} \geq \text{IL}_{peak} = 5.31\text{A} \rightarrow 15\text{A}$ ✓
 $\text{ID}_{rms} = \text{IL}_{avg} \times \sqrt{D} = 2.72\text{A}$
 $\text{P}_{cond} = \text{ID}_{rms}^2 \times R_{ds(on)} \times 1.3 = 2.72^2 \times 0.637 = 4.71\text{W}$ (30% derating)
 $\text{P}_{sw} = 0.5 \times V_{ds} \times \text{IL}_{peak} \times (\tau_{on} + \tau_{off}) \times f_s = 0.5 \times 36 \times 5.31 \times 16\text{ns} \times 300\text{kHz} = 0.46\text{W}$
 $\text{P}_{total} = 4.71 + 0.57 = 5.12\text{W}$

DIODE — STPS10L60S (60V, 10A, $V_f = 0.4\text{V}$, DPAK)

$V_f_stress = \text{Vin} + \text{Vout} = 36\text{V}$
 $V_f_rated = 36 \times 1.2 = 43.2\text{V} \rightarrow 60\text{V}$ ✓
 $\text{ID}_{avg} = \text{Iout} = 3\text{A}$
 $\text{ID}_{peak} = \text{IL}_{peak} = 5.31\text{A}$
 $\text{P}_{diode} = V_f \times \text{Iout} \times (1-D) = 0.4 \times 3 \times 0.667 = 0.80\text{W}$

OUTPUT CAPACITOR — KRM55WR71H476MH13K x2 (47uF, 50V, X7R)

$C_{min} = \text{Iout} \times D / (f_s \times \Delta\text{Vout}) = 27.7\mu\text{F}$
 $C_{nominal} = 2 \times 47 = 94\mu\text{F}$
 $C_{derated} @ 12\text{V} = 2 \times 39.6 = 79.2\mu\text{F} > 27.7\mu\text{F}$ ✓
 $\Delta\text{Vout_target} = 1\% \times 12 = 120\text{mV}$
 $\text{ESR}_{max} = 0.5 \times \Delta\text{Vout} / \Delta\text{IL} = 49.6\text{m}\Omega$
(the ESR can contribute max by a 50% to the total allowed ripple)
 $\text{ESR}_{eff} = 4\text{m}\Omega / 2 = 2\text{m}\Omega \ll 49.6\text{m}\Omega$ ✓
 $\text{IC}_{rms} = \text{Iout} \times \sqrt{(D/D')} = 2.12\text{A}$
 $\text{IC}_{rms/cap} = 2.12 / 2 = 1.06\text{A}$ ✓

INPUT CAPACITOR — KRM55WR71H476MH13K x2 (47uF, 50V, X7R)

$C_{min} = \text{Iout} \times D / (f_s \times \Delta\text{Vin} \times D') = 20.8\mu\text{F}$
 $C_{nominal} = 2 \times 47 = 94\mu\text{F}$
 $C_{derated} @ 24\text{V} = 2 \times 26.6\mu\text{F} = 53.2\mu\text{F} > 20.8\mu\text{F}$ ✓
 $\Delta\text{Vin_target} = 1\% \times 24 = 240\text{mV}$
 $\text{ESR}_{max} = 0.5 \times \Delta\text{Vin} / \text{IL}_{peak} = 22.6\text{m}\Omega$
 $\text{ICin}_{rms} = \text{IL} \times \sqrt{(D \times D')} = 2.22\text{A}$
 $\text{Vrated} = 32 \times 1.5 = 48\text{V} \rightarrow 50\text{V}$ ✓
 $\text{Cin impedance } Z_{cin} = 1 / (2\pi \times f_s \times \text{Cin}) + \text{ESRcin} = 0.413\Omega$
 $\text{Cdamp impedance } Z_{cdamp} = 1 / (2\pi \times f_s \times \text{Cdamp}) + \text{ESRdamp} = 0.013\Omega$
 $\text{IRMS Total Input Caps } (\text{Cin} + \text{Cdamp}) = \text{IL} \times \sqrt{(D \times D')} = 2.22\text{A}$
 $\text{IRMS Cdamp} = Z_{cdamp} / (Z_{cdamp} + Z_{cin}) = 2.16\text{A}$ ✓

POWER BUDGET

$\text{P}_{mosfet} = 5.17\text{W}$
 $\text{P}_{diode} = 0.80\text{W}$
 $\text{P}_{total} = 6.08\text{W}$
 $\eta_{estimated} = \text{Pout} / (\text{Pout} + \text{Ploss}) = 36 / (36 + 6.08) = 85.6\%$

MOSFET: FQD18N20V2
Driver: TLP5702(E)
 $\text{VDRV} = 24\text{V}$

Method 1 – Total Gate Charge

$\text{Qg} = 20\text{nC}$
 $\text{Idrv,max} = 2.5\text{A}$

$\tau_r = \text{Qg} / \text{Idrv,max} = 8\text{ns}$
 $\tau_f = \text{Qg} / \text{Idrv,max} = 8\text{ns}$

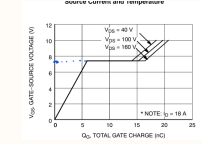
Method 2 – Miller Charge :

$\text{Qgd} \approx 7\text{nC}$
 $\text{Vplateau} \approx 7.8\text{V}$

$\text{Ig,on} = \min[(\text{VDRV} - \text{Vplateau}) / \text{Rgon}, \text{Idrv,max}]$
 $\text{Ig,off} = \min[(\text{VDRV} - \text{Vplateau}) / \text{Rgoff}, \text{Idrv,max}]$

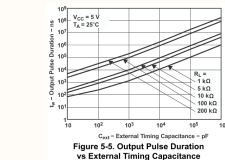
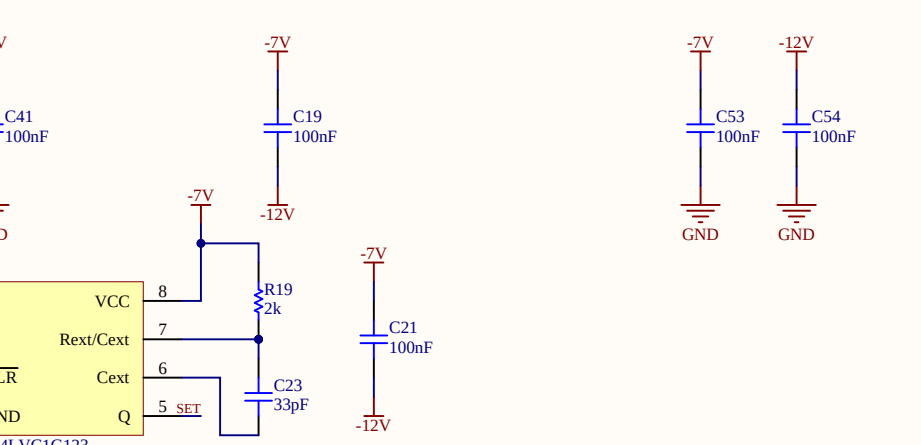
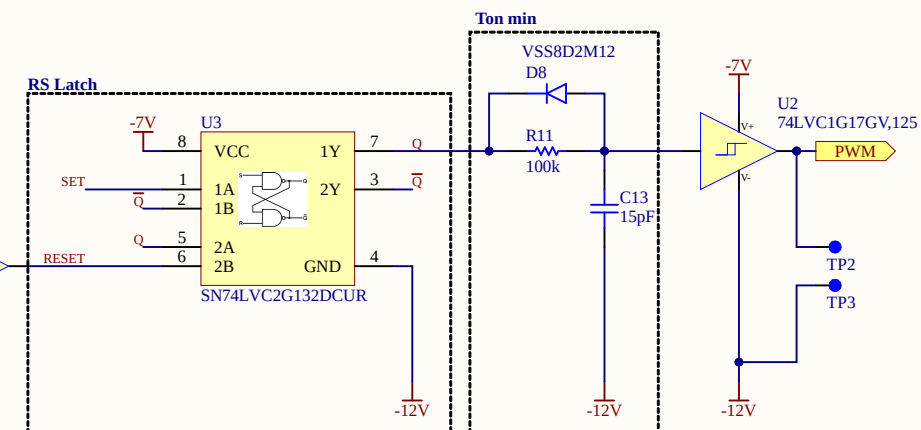
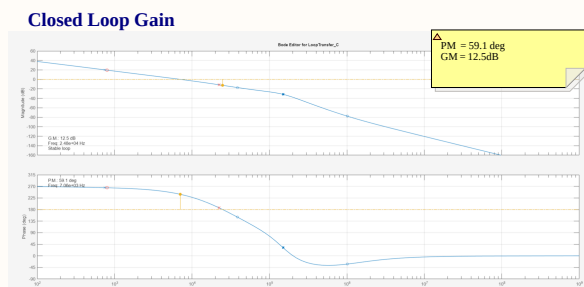
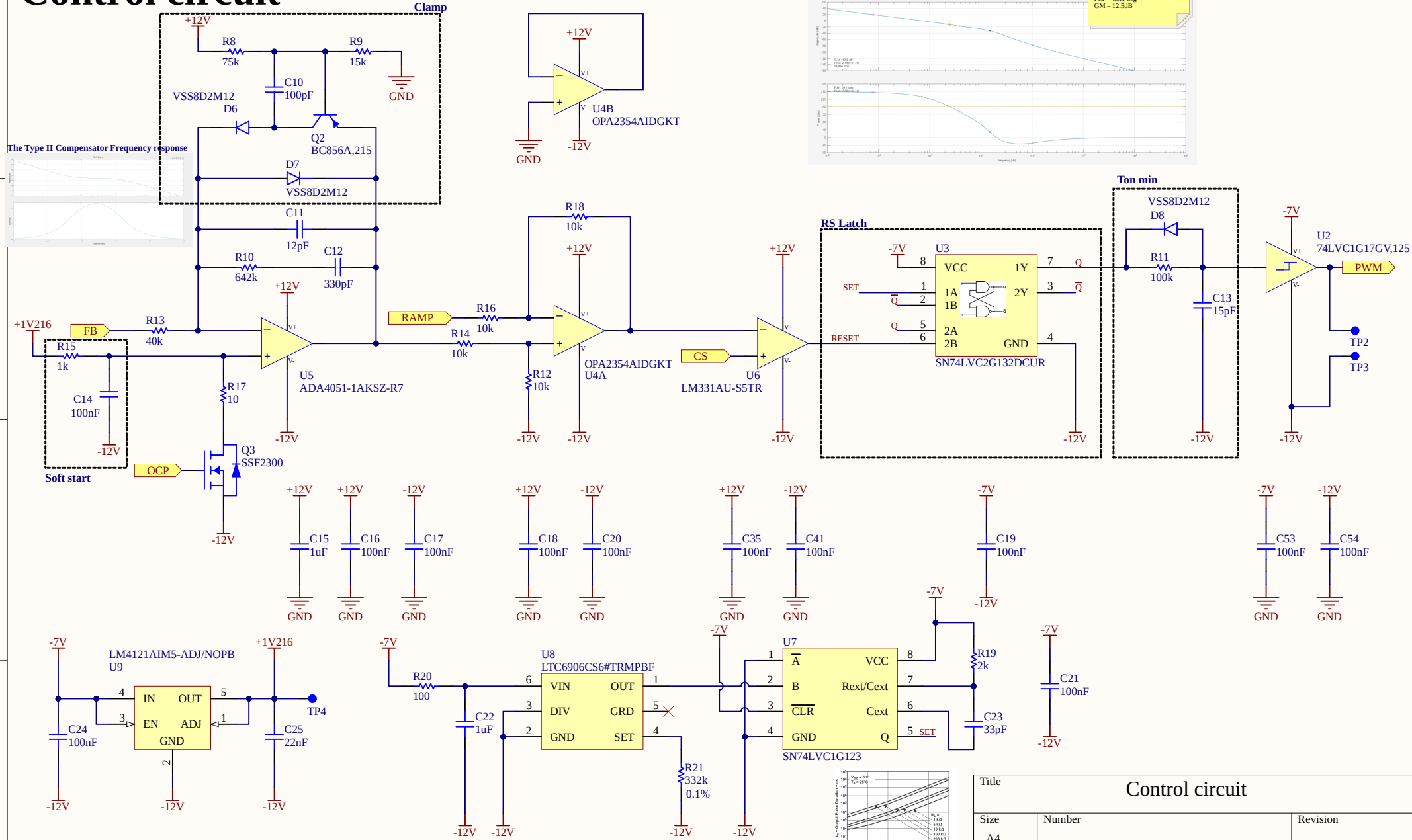
$\tau_r = \text{Qgd} / \text{Ig,on} \approx 2.8\text{ns}$
 $\tau_f = \text{Qgd} / \text{Ig,off} \approx 2.8\text{ns}$

Switching Loss:
 $\text{Psw} = 0.5 \times \text{VDS} \times \text{ID} \times (\tau_r + \tau_f) \times f_s$



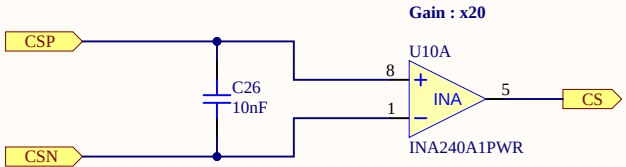
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Control circuit

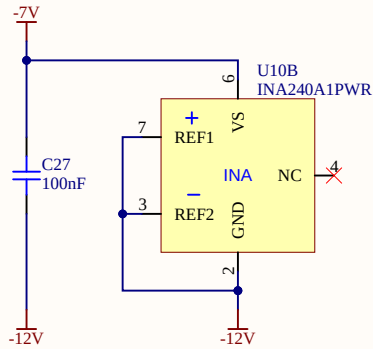


Control circuit		
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Current sense



^A Total current gain = 0.015 x 20 = 0.3



Title		
Current sense		
Size	Number	Revision
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Date:	6/28/2026	Sheet of
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Input switch

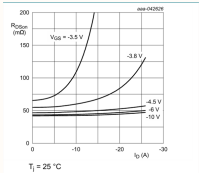
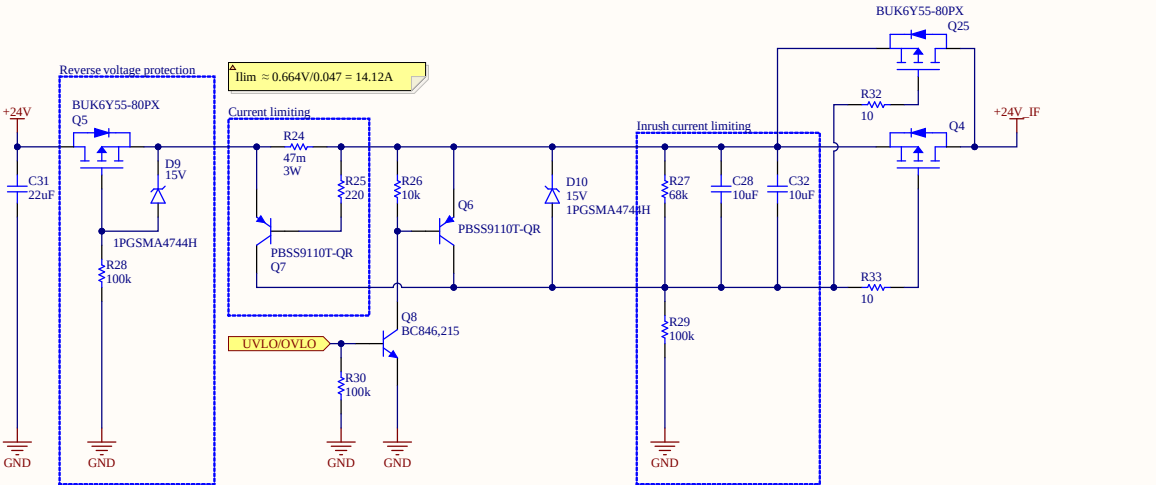


Fig. 7. Drain-source on-state resistance as a function of drain current; typical values

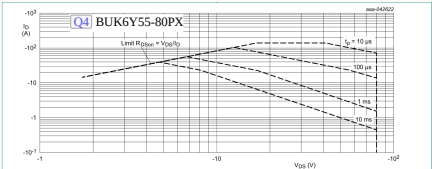


Fig. 3. Safe operating area; junction to mounting base; continuous and peak drain currents as a function of drain-source voltage

INRUSH CURRENT SIZING — 300uF Load, Vin_max=32.5V

Load = Cf + Cdamp + Cin = 5.45u + 53.3u + 220u = 278.75u ≈ 300u
I = C × dV/dt → Iinrush = Ctotal × Vin_max / t_on
Target: 1.3× SOA margin at derated TJ=66.7°C

ISOA(TJ) = ISOA(25°C) × (TJ_ABSMAX - TJ) / (TJ_ABSMAX - 25°C) = ISOA(25°C) × 0.722
[1] Iinrush(t) = (300uF × 32.5V) / t = 9.75m / t
[2] ISOA_derated(t) = 8 × (t)^(-0.602) × 0.722 / 1.3

[1] = [2] → 9.75m / t = 4.44 × t^(-0.602)
→ 9.75m = 4.44 × t^(0.398)
→ t_on = 8.34ms
Result: t_on ≈ 8.3ms → Iinrush ≈ 1.2A

RECOMMENDED: Iinrush_max = 1.2A (t_on=8.3ms)
Provides 1.3× SOA margin at worst-case OVLO=32.5V and derated TJ=66.7°C.

Vgate(steady state) = Vin × R29 / (R27 + R29) = 24 × 100k / (68k + 100k) = 24 × 0.595 = 14.3V
VGS,ss = Vgate,ss - Vin = 14.3 - 24 = 9.7V more than Vgs threshold and less than Vgs max of the Mosfet

BUK6Y55-80P [Q4] — STEADY-STATE THERMAL & SOA VERIFICATION

SOA evaluated @ Vds = Vin_max = OVLO threshold = 32.5V
ILOAD evaluated @ Vin_min = 18V (worst project ambient spec)
TA,MAX = 40°C (project ambient spec)
ILOAD,MAX = Pout / (η × Vin_min) = 36 / (0.88 × 18) = 2.27A

RthCA — DERIVED FROM RthJA (LPAK56/Power-SO8 package)

RthCA = RthJA - RthJC
With less than 6cm² copper pour (minimal footprint):
RthJA = 91°C/W (max) | RthJC(j-mb) = 1.0°C/W = 91 - 1.0 = 90°C/W
NOTE: RthJA taken from reference LPAK56/Power-SO8 package datasheet as conservative stand-in. BUK6Y55-80P datasheet only specifies Rth(j-mb)=1°C/W, not Rth(j-a). Used here: RthCA = 90°C/W (with less than 6cm² copper pour)

RDSon DETERMINATION From Fig.7 (BUK6Y55-80P datasheet)

Gate drive: Vgs_ss = -9.7V → Use Vgs = -10V curve (closest, conservative)
At ILOAD,MAX = 2.27A, Vgs=-10V, TJ=25°C:
RDSon(25°C) = 42mOhm (read from Fig.7 datasheet curve)

NOTE: Datasheet MAX RDSon(25°C)=55mOhm used below for worst-case design margin.
RDSon(125°C) = a_ratio × RDSon(25°C)_max = 1.85 × 55mOhm = 101.75mOhm

STEP 1 — SOA AT t=8.34ms (Vds=OVLO=32.5V, not on datasheet graph)

Read SOA curve (Fig.3, BUK6Y55-80P datasheet)
@ 25°C, Vds=32.5V at two known points:
t1 = 1ms → ISOA(t1) = 8A t2 = 10ms → ISOA(t2) = 2A

ISOA(t) = a × t^m
m = ln(ISOA(t1)/ISOA(t2)) / ln(t1/t2) a = ISOA(t1) / t1^m
= ln(8/2) / ln(1ms/10ms) = 8 / (1e-3)^(-0.602)
= 1.386 / -2.303 = 8 / 63.97
= -0.602 = 0.125 A·ms

ISOA(8.34ms, 25°C) = a × (8.34ms)^m = 0.125 × (8.34)^(-0.602) = 0.125 × 16.2 = 2.03A

STEP 2 — TC AND TJ CALCULATION

Tc = TA,MAX + RthCA × ILOAD,MAX² × RDSon(TJ)
= 40 + 90 × (2.27)² × 101.75e-3 = 87.2°C
TJ = Tc + Rth(j-mb) × [ILOAD,MAX² × RDSon(TJ)]
= 87.2 + 1.0 × (2.27² × 101.75) = 87.7°C
RESULT — STEADY STATE

TJ = 87.7°C < TJ,max = 175°C
Thermal margin = 87.3°

STEP 3 — DERATE SOA TO ACTUAL TJ AND RECHECK

ISOA(TJ) = ISOA(25°C) × (TJ_ABSMAX - TJ) / (TJ_ABSMAX - 25°C)
ISOA(8.34ms, 87.7°C) = 2.03 × (175-87.7) / (175-25) = 1.18A

STEP 4 — INRUSH SOA CHECK (Iinrush=1.17A, t=8.34ms)

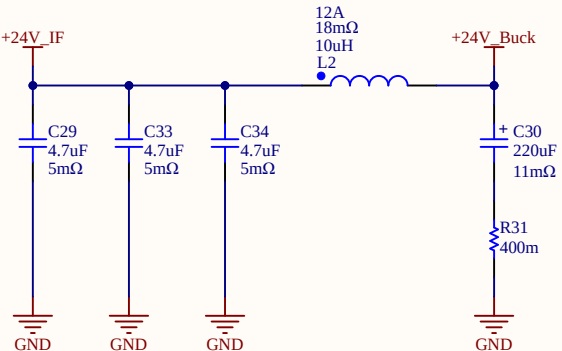
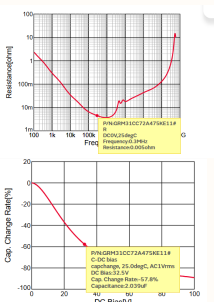
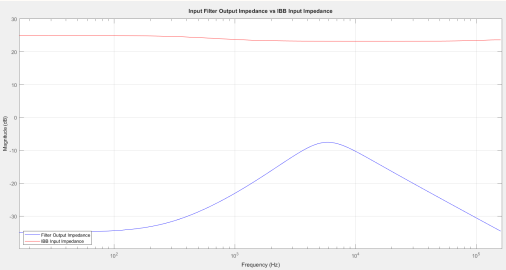
CHECK: Iinrush = 1.17A < ISOA_derated = 1.18A ✓ (marginal)
SOA margin = 1.18/1.17 = 1.01×

Δ Margin ≈ 1.0× — INSUFFICIENT (target ≥ 1.3×) With less than 6cm² copper (RthCA=90°C/W), a single BUK6Y55-80P barely meets SOA at the inrush point.

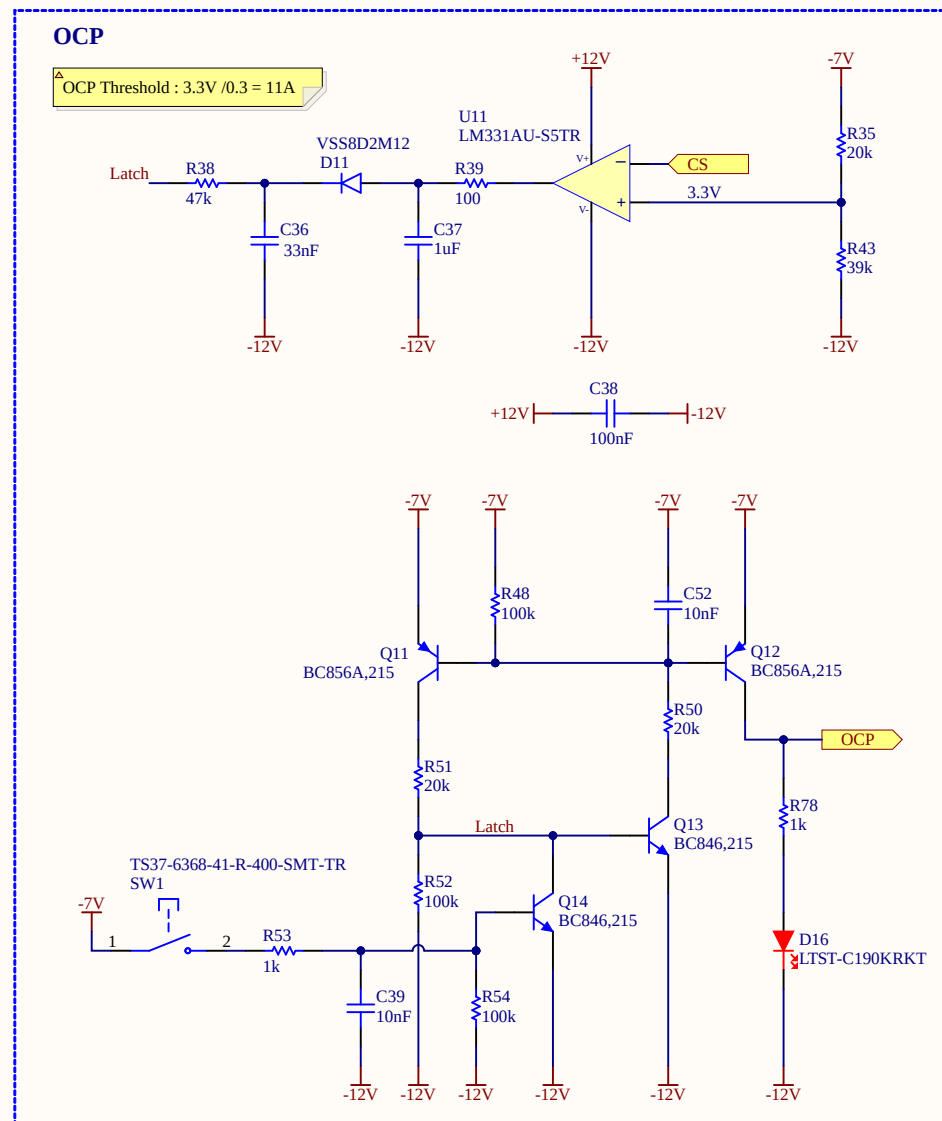
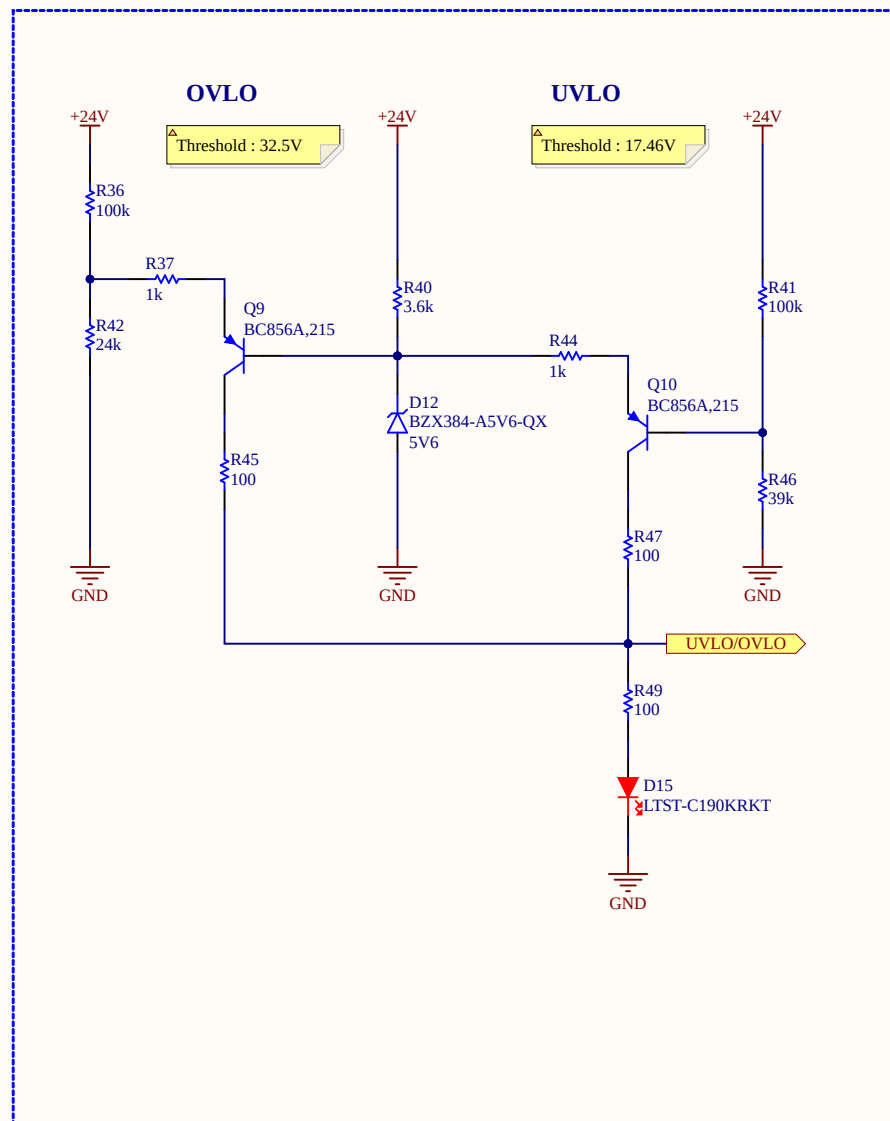
→ USE 2 MOSFETs IN PARALLEL to restore safe margin. Each device then carries ~half the current:
ILOAD,perFET = 1.135A, Iinrush,perFET = 0.585A
TJ,perFET ≈ 51.9°C → ISOA_derated ≈ 1.67A
SOA margin = 1.67/0.585 = 2.85× ✓

Title			Input switch
Size	Number	Revision	
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File:	C:\Users\...Input switch.SchDoc	Drawn By:	Mohamed Drissi

^ INPUT EMI FILTER : Design Notes		
Standard: EN 55032:2015 Class B — Conducted Emissions		
Limit @ 300kHz = 56 dBμV (quasi-peak)		
Frequency range: 150kHz – 30MHz		
SPECIFICATIONS:		
Vin = 24V Vout = -12V Iout = 3A fs = 300kHz D = 0.333 Cin = 53.2uF (derated @24V)		
STEP 1 : NOISE LEVEL ESTIMATE		
First harmonic amplitude:		
I1 = (2×IL×sin(π×D)) / π = (2×4.705×sin(π×0.333)) / π = 2.60A		
Noise voltage across Cin at fs:		
Vnoise = I1 / (2π×fs×Cin) = 2.60 / (2π×300k×53.2u) = 25.9mV = 25.9×10³ μV		
Estimated noise level:		
Noise dBμV = 20log(25.9×10³) = 88.3 dBμV		
Required attenuation:		
Att = Noise dBμV - Vmax = 88.3 - 56 = 32.3 dB		
Adding 10dB safety margin:		
Att_target = 32.3 + 10 = 42.3 dB		
STEP 2 : INDUCTOR SELECTION		
Lf = 10uH → Selected: 10uH, 12A, SMD		
STEP 3 : CAPACITOR SELECTION		
Resonance one decade below fs:		
Cf = Cin / (Cin×Lf×(2π×fs/10)² - 1) = 53.2u / (53.2u×10u×(2π×30k)² - 1) = 3μF		
Required attenuation:		
Cf= 1/Lf × (10^(Att/40)/(2π×fs))² = 5μF		
Select higher value: Cf = max(5μF,3μF) = 5μF→ 3x 4.7μF selected derated to 2uF @Vin,max = 32.5V		
NOTE: Cf is a SEPARATE dedicated filter capacitor distinct from Cin.		
fc_filter = 1/(2π×√(Lf×Cf)) = 1/(2π×√(10u×6μF)) = 20.5kHz		
Attenuation verification:		
A = 40×log(fs/fc_filter) = 40×log(300k/20.5k) = 46.6dB >> 42.3dB ✓		
STEP 4 : DAMPING CAPACITOR		
Cd ≥ 4×Cin = 4×53.2u = 212.8uF → 220uF selected		
Required ESRd:		
ESRd ≤ √(Lf/Cin) = √(10u/53.2u) = 0.434Ω		
Damping resistor power:		
P_Rdamp ≈ 0.5×Δlin²×Rdamp = 0.5×(1.21)²×1.5 = 0.8W		
Selected Cd: 220uF 35V polymer electrolytic ESR_polymer ≈ 11mΩ < 434mΩ ✓		
NOTE: Polymer ESR too low, added series resistor Rd = 0.4Ω in series with Cd to meet ESRd target		
Voltage rating: Vin_max×1.25 = 32×1.25 = 40V → 50V ✓		
Cin impedance Zcin = 1/(2π×fs×Cin) + ESRcin = 0.413Ω		
Cdamp impedance Zcdamp = 1/(2π×fs×Cdamp) + ESRdamp = 0.013Ω		
IRMS Total Input Caps (Cin +Cdamp) = IL×√(D×D') = 2.22A		
IRMS Cdamp (current divider) = Zcin / (Zcdamp +Zcin) = 0.062A✓		
STEP 4 : MIDDLEBROOK CRITERION		
Zin_conv = Vin²×η/Pout = 576×0.88/36 = 14.1Ω		
Full frequency spectrum verified using matlab: see Plot		
EMC STANDARD : EN 55032:2015		
EN 55032 "Electromagnetic Compatibility of Multimedia Equipment" . It defines conducted emission limits on the power port for Class B equipment (residential environment):		
150kHz – 500kHz : 56–66 dBμV (quasi-peak)		
46–56 dBμV (average)		
500kHz – 30MHz : 56 dBμV (quasi-peak)		
46 dBμV (average)		
Worst case for this design at fs=300kHz: 56 dBμV QP.		
Reference : Conducted EMI Characteristics And Mitigation Technique		
https://www.ti.com/lit/an/snva489c/snva489c.pdf		

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Input filter																				
																				
																				
																				
<table><tr><th colspan="3">Title</th></tr><tr><td colspan="3">Input filter</td></tr><tr><th>Size</th><th>Number</th><th>Revision</th></tr><tr><td>A4</td><td></td><td></td></tr><tr><td>Date:</td><td>6/28/2026</td><td>Sheet of</td></tr><tr><td>File:</td><td>C:\Users\...\Input filter.SchDoc</td><td>Drawn By: Mohamed Drissi</td></tr></table>			Title			Input filter			Size	Number	Revision	A4			Date:	6/28/2026	Sheet of	File:	C:\Users\...\Input filter.SchDoc	Drawn By: Mohamed Drissi
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Protections



Title		
Protections		
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A4		
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DATASHEET (Toshiba 2SC6076):
V_{CEO} = 80V | V_{CB0} = 160V | I_C = 3A DC / 5A pulse P_c = 10W @ T_c=25°C | T_J_max = 150°C
V_{CE}(sat) = 0.5V max @ I_C=1A | h_{FE} = 100-450
Thermal resistances (large copper): R_{th}(c-a) = 21.2 °C/W R_{th}(j-c) = 2 °C/W (NOTE: R_{th} values borrowed from ROHM AN-64-AN140E (TO-252, JEDEC 2s2p): θJA=23.2, ΨJT~2°C/W → R_{th}(j-c)~2, R_{th}(c-a)=21.2. 25C6076 datasheet gives no R_{th}; same-package values used as stand-in. Assumes large copper pour — less copper → higher T_j)
Linear-regulator operation = CONTINUOUS → use DC SOA.

RAIL 1 — +12V REGULATOR (Q15 // Q16)

OPERATING POINT (per device, current shared):

V_{ce} = V_{in} - V_{out} = 24 - 12 = 12V

I_c_total = I_{lim} = 500mA

I_c_perdevice = 500mA / 2 = 250mA

POWER DISSIPATION (per device):

P_d = V_{ce} × I_c_perdevice = 12 × 0.25 = 3.0W (< P_c=10W @ T_c=25°C ✓)

SOA CHECK (DC operation, continuous):

Operating point: V_{ce}=12V, I_c=250mA

DC SOA curve @ V_{ce}=12V :

I_c_SOA(DC, 12V, 25°C) ≈ 0.8A per device

THERMAL:

R_{th}(c-a) = 21.2 °C/W | R_{th}(j-c) = 2 °C/W T_c = T_A + R_{th}(c-a) × P_d = 40 + 21.2 × 3.0 = 103.6 °C T_j = T_c + R_{th}(j-c) × P_d = 103.6 + 2 × 3.0 = 109.6 °C ✓ < 150°C

Derated SOA to T_c:

I_c_SOA(DC, T_c) = 0.8 × (150-T_c)/(150-25)

= 0.8 × (150-103.6)/(150-25)

= 0.297A

CHECK: I_c_perdevice = 0.25A < I_c_SOA_derated = 0.297A ✓

SOA margin = 0.297/0.25 = 1.19×

RAIL 2 — -7V REGULATOR (Q20 // Q21) — 2 PARALLEL BJTs

OPERATING POINT (per device, current shared 2-way):

V_{ce} = V_{in} - V_{out} = 12 - (-7) = 19V

I_c_total = I_{lim} = 318mA

I_c_perdevice = 318mA / 2 = 159mA

POWER DISSIPATION (per device):

P_d = V_{ce} × I_c_perdevice = 19 × 0.159 = 3.021W (< P_c=10W ✓)

SOA CHECK (DC operation, continuous):

Operating point: V_{ce}=19V, I_c=159mA

DC SOA curve @ V_{ce}=19V: I_c_SOA(DC, 19V, 25°C) ≈ 0.3A per device

THERMAL:

R_{th}(c-a) = 21.2 °C/W | R_{th}(j-c) = 2 °C/W

T_c = T_A + R_{th}(c-a) × P_d = 40 + 21.2 × 3.021 = 104 °C

T_j = T_c + R_{th}(j-c) × P_d = 104 + 2 × 3.021 = 110.1 °C ✓ < 150°C

Derated SOA to T_c:

I_c_SOA(DC, T_c) = 0.3 × (150-T_c)/(150-25)

= 0.3 × 46.0/125

= 0.110A

CHECK: I_c_perdevice = 0.159A > I_c_SOA_derated = 0.110A ✗ FAIL

SOA margin = 0.110/0.159 = 0.69×

CONCLUSION (2 BJTs): FAIL

Thermal passes (T_j=110°C<150°C) but DC SOA at V_{ce}=19V

SOLUTIONS:

(1) Add a 3rd parallel BJT → splits current further

(2) Pre-regulator/series dropper to lower V_{ce} (moves left on SOA, away from second breakdown)

(3) Wider-SOA pass transistor

→ Adopt solution (1): add 3rd parallel device.

RAIL 2 — -7V REGULATOR (Q20 // Q21 // Q26) — 3 PARALLEL BJTs

OPERATING POINT (per device, current shared 3-way):

V_{ce} = 19V

I_c_total = I_{lim} = 318mA

I_c_perdevice = 318mA / 3 = 106mA

POWER DISSIPATION (per device):

P_d = V_{ce} × I_c_perdevice = 19 × 0.106 = 2.014W (< P_c=10W ✓)

SOA CHECK (DC operation, continuous):

Operating point: V_{ce}=19V, I_c=106mA

DC SOA curve @ V_{ce}=19V:

I_c_SOA(DC, 19V, 25°C) ≈ 0.3A per device

THERMAL:

R_{th}(c-a) = 21.2 °C/W | R_{th}(j-c) = 2 °C/W

T_c = T_A + R_{th}(c-a) × P_d = 40 + 21.2 × 2.014 = 82.7 °C

T_j = T_c + R_{th}(j-c) × P_d = 82.7 + 2 × 2.014 = 86.7 °C ✓ < 150°C

Derated SOA to T_c:

I_c_SOA(DC, T_c) = 0.3 × (150-T_c)/(150-25) = 0.3 × 67.3/125 = 0.162A

CHECK: I_c_perdevice = 0.106A < I_c_SOA_derated = 0.162A ✓ PASS

SOA margin = 0.162/0.106 = 1.52×

CONCLUSION (3 BJTs): PASS

3 parallel devices bring per-device current to 106mA,

T_j=86.7°C, SOA margin 1.52× at V_{ce}=19V. ✓

EMITTER BALLAST — CURRENT SHARING

RE = 1Ω (R55/R56 and R66/R67)

Purpose: forces current sharing between parallel devices, prevents thermal runaway from VBE mismatch.

Voltage drop across RE:

Rail1: V_{RE} = 0.25A × 1 = 250mV

Rail2: V_{RE} = 0.159A × 1 = 159mV

V_{RE} ≥ 5×ΔV_{BE} rule (ΔV_{BE}≈10mV typical mismatch (between the 2 transistors in //):

5×10mV = 50mV → Rail1 (250mV) ✓ | Rail2 (159mV) marginal

PRE: Rail1 = 0.25×1 = 250mW → 0.500W rated ✓

Linear regulators

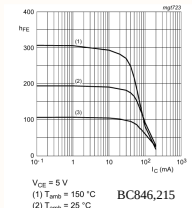
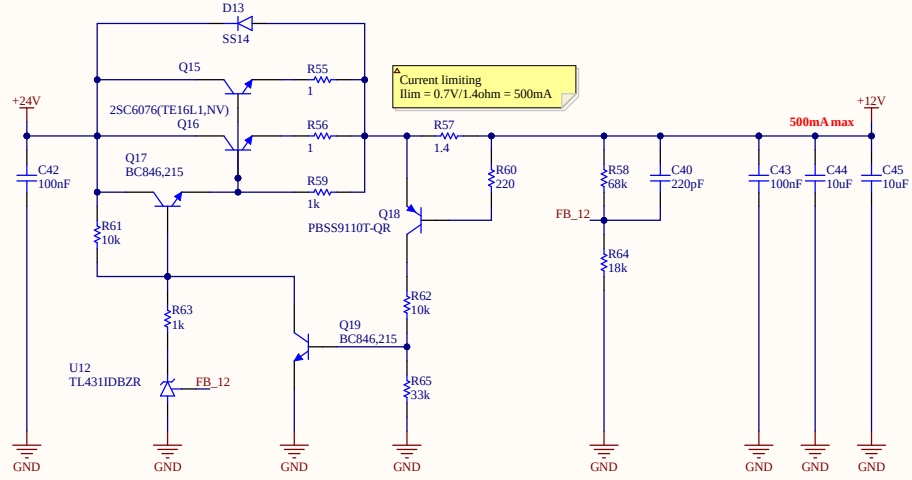
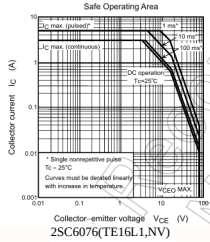


fig. 2. Group A: DC current gain as a function of collector current; typical values



⚠ To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable h_{FE}. When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet The base-emitter resistor (and **R69**) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.

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Linear regulators		
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